

Project Report: Asian Option Pricing

Pricing Asian Options in the Black-Scholes Model

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1 General Setup and Notation

Throughout this report, we work on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ satisfying the usual conditions. We fix a risk-neutral pricing measure $\mathbb{Q}$ and denote by $W = (W_t)_{t \geq 0}$ a standard Brownian motion under $\mathbb{Q}$.

Definition 1.1 (Black-Scholes parameters). The following parameters are fixed in Section 2: $S_0 > 0$ (initial asset price), $r \geq 0$ (risk-free rate), $\sigma > 0$ (volatility), $T > 0$ (maturity), $K > 0$ (strike), $n \in \mathbb{N}$ (discretisation steps), and $M \in \mathbb{N}$ (Monte Carlo sample size).

Definition 1.2 (Black-Scholes dynamics). Under $\mathbb{Q}$, the asset price $S = (S_t)_{t \geq 0}$ satisfies

$$dS_t = rS_t dt + \sigma S_t dW_t, \quad S_0 > 0. \quad (1)$$

By Itô's formula the unique strong solution is

$$S_t = S_0 \exp\left(\sigma W_t + \left(r - \frac{\sigma^2}{2}\right)t\right). \quad (2)$$

2 Pricing Asian Options in the Black-Scholes Model

This section follows [8], Sections 3–4.

2.1 The Asian Option

Definition 2.1 (Asian call option). An *Asian call option* with maturity T and strike K has payoff

$$\Psi(A_T) = (A_T - K)^+, \quad (3)$$

where $A_T = \frac{1}{T} \int_0^T S_t dt$ is the continuous-time arithmetic average of the asset over $[0, T]$.

Definition 2.2 (Discretised averages). On the uniform grid $t_i = iT/n$, $i = 0, \dots, n$, we approximate A_T by

$$A_T^{\text{arith}} := \frac{1}{n+1} \sum_{i=0}^n S_{t_i}, \quad (4)$$

$$A_T^{\text{geom}} := \exp\left(\frac{1}{n+1} \sum_{i=0}^n \ln S_{t_i}\right). \quad (5)$$

Definition 2.3 (Asian call price). The no-arbitrage price at time 0 is

$$\text{CallAsian}(r, \sigma, S_0, T, K) = e^{-rT} \mathbb{E}^{\mathbb{Q}}[(A_T - K)^+]. \quad (6)$$

Remark 2.4. Unlike the geometric case, no closed-form price exists for the arithmetic Asian option in the Black-Scholes model, since the arithmetic average of log-normals is not log-normal. This motivates Monte Carlo methods.

2.2 Question 1 — Black-Scholes Formula and Convexity

Theorem 2.5 (Black-Scholes formula). *The price at time 0 of a European call with strike K and maturity T is*

$$\text{CallBS}(S_0, K, r, \sigma, T) = S_0 \Phi(d_1) - K e^{-rT} \Phi(d_2), \quad (7)$$

where Φ is the standard normal CDF and

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}. \quad (8)$$

Property 2.6 (Convexity in K). The map $K \mapsto \text{CallBS}(S_0, K, r, \sigma, T)$ is convex and decreasing on $(0, \infty)$.

Proof. Convexity follows because $K \mapsto (S_T - K)^+$ is convex for every realisation of S_T and the discounted expectation preserves convexity. Analytically, $\partial^2 \text{CallBS} / \partial K^2 = e^{-rT} \phi(d_2) / (\sigma \sqrt{T} K) > 0$, where ϕ denotes the standard normal PDF. $\square$

Example 2.7 (Numerical convexity check). With $S_0 = 50$, $r = 0.05$, $\sigma = 0.30$, $T = 1$ and a uniform strike grid $K \in \{35, 36, \dots, 55\}$ ($h = 1$), we compute $\Delta_h^2 C_{\text{BS}}(K) = C_{\text{BS}}(K - 1) - 2C_{\text{BS}}(K) + C_{\text{BS}}(K + 1)$ at each interior point. Table 1 reports the call prices; the minimum second finite difference is $1.893 \times 10^{-2} > 0$, confirming Property 2.6 numerically.

Table 1: Black-Scholes call prices as a function of strike K ($S_0 = 50$, $r = 0.05$, $\sigma = 0.30$, $T = 1$). All second finite differences are positive, confirming convexity.

K	C_{BS}	K	C_{BS}	K	C_{BS}
35	17.1977	42	11.8022	49	7.6103
36	16.3638	43	11.1252	50	7.1156
37	15.5490	44	10.4740	51	6.6463
38	14.7544	45	9.8487	52	6.2016
39	13.9814	46	9.2497	53	5.7812
40	13.2310	47	8.6771	54	5.3842
41	12.5044	48	8.1307	55	5.0100

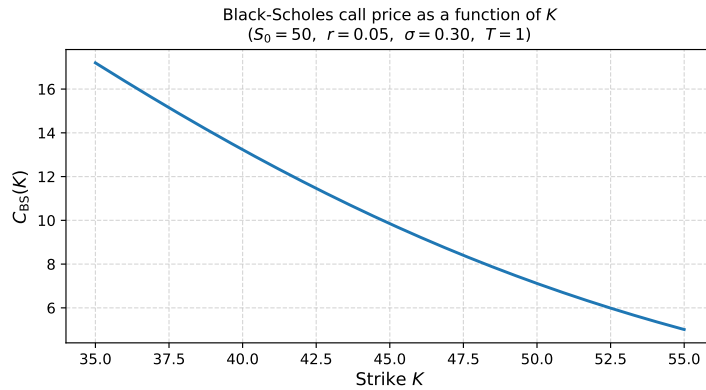


Figure 1: Black-Scholes call price as a function of strike K ($S_0 = 50$, $r = 0.05$, $\sigma = 0.30$, $T = 1$). The convex shape is clearly visible.

2.3 Question 2 — Monte Carlo Pricing

Definition 2.8 (Monte Carlo estimator). Given M i.i.d. copies $X^{(1)}, \dots, X^{(M)}$ of a random variable X , the Monte Carlo estimator of $\mathbb{E}[X]$ is

$$\bar{X}_M := \frac{1}{M} \sum_{j=1}^M X^{(j)} \xrightarrow{a.s.} \mathbb{E}[X] \quad \text{as } M \rightarrow \infty. \quad (9)$$

Remark 2.9 (Exact path simulation). We simulate the asset using the exact log-normal increment:

$$S_{t_{i+1}} = S_{t_i} \exp\left(\left(r - \frac{\sigma^2}{2}\right)\Delta t + \sigma\sqrt{\Delta t} Z_i\right), \quad Z_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1),$$

with $\Delta t = T/n$. This introduces no time-discretisation error.

We set $S_0 = 50$, $n = 50$, $M = 200,000$ and explore the parameter combinations in Table 2.

Table 2: Parameter grid for numerical experiments.

r	σ	K	T
0.00	0.2, 0.3, 0.4	35, 40, 45, 50, 55	1, 2
0.03	0.2	35, 40, 45, 50, 55	1, 2
0.04	0.4	35, 40, 45, 50, 55	1, 2
0.05	0.3	35, 40, 45, 50, 55	1, 2
1.00	0.1	35, 40, 45, 50, 55	1, 2

Selected results are reported in Tables 3 and 4.

Table 3: Monte Carlo Asian call prices, 95% CI widths, and European call for $T = 1$ ($S_0 = 50$, $n = 50$, $M = 200,000$).

r	σ	T	K	$\hat{C}^a$	CI ^a width	$\hat{C}^g$	CI ^g width	C_{BS}
0.00	0.20	1	35	14.9963	0.0503	14.8343	0.0503	15.1241
0.00	0.20	1	40	10.0324	0.0497	9.8807	0.0493	10.5930
0.00	0.20	1	45	5.4967	0.0444	5.3587	0.0435	6.7946
0.00	0.20	1	50	2.2713	0.0318	2.1882	0.0308	3.9828
0.00	0.20	1	55	0.6996	0.0180	0.6605	0.0172	2.1460
0.00	0.30	1	35	15.0229	0.0760	14.6895	0.0746	15.7148
0.00	0.30	1	40	10.3249	0.0723	10.0059	0.0703	11.7672
0.00	0.30	1	45	6.2831	0.0631	6.0780	0.0613	8.5064
0.00	0.30	1	50	3.4255	0.0499	3.2337	0.0474	5.9618
0.00	0.30	1	55	1.6691	0.0361	1.5115	0.0332	4.0705
0.00	0.40	1	35	15.2222	0.1002	14.6085	0.0968	16.6856
0.00	0.40	1	40	10.8515	0.0940	10.3092	0.0895	13.1956
0.00	0.40	1	45	7.2349	0.0830	6.8127	0.0785	10.2856
0.00	0.40	1	50	4.5425	0.0694	4.1996	0.0644	7.9260
0.00	0.40	1	55	2.7807	0.0562	2.4895	0.0509	6.0541
0.03	0.20	1	35	15.2968	0.0503	15.1217	0.0495	16.1176
0.03	0.20	1	40	10.4816	0.0496	10.3120	0.0491	11.6120
0.03	0.20	1	45	5.9792	0.0451	5.8432	0.0442	7.7146
0.03	0.20	1	50	2.6376	0.0339	2.5280	0.0325	4.7067
0.03	0.20	1	55	0.8649	0.0200	0.7984	0.0187	2.6467
0.04	0.40	1	35	15.5172	0.0992	14.9517	0.0958	17.7432
0.04	0.40	1	40	11.2543	0.0933	10.7392	0.0893	14.2278
0.04	0.40	1	45	7.7028	0.0838	7.2491	0.0793	11.2473
0.04	0.40	1	50	4.9283	0.0712	4.5820	0.0664	8.7891
0.04	0.40	1	55	3.0471	0.0583	2.7293	0.0524	6.8064
0.05	0.30	1	35	15.5281	0.0747	15.1321	0.0730	17.1977
0.05	0.30	1	40	10.9611	0.0720	10.5904	0.0697	13.2310
0.05	0.30	1	45	6.9756	0.0647	6.6944	0.0621	9.8487
0.05	0.30	1	50	3.9573	0.0527	3.7220	0.0497	7.1156
0.05	0.30	1	55	2.0245	0.0390	1.8315	0.0358	5.0100
1.00	0.10	1	35	18.7820	0.0180	17.4268	0.0152	37.1242
1.00	0.10	1	40	16.9376	0.0180	15.5768	0.0153	35.2848
1.00	0.10	1	45	15.1079	0.0180	13.7481	0.0153	33.4454
1.00	0.10	1	50	13.2573	0.0181	11.9088	0.0152	31.6060
1.00	0.10	1	55	11.4280	0.0181	10.0689	0.0153	29.7666

Table 4: Monte Carlo Asian call prices, 95% CI widths, and European call for $T = 2$ ($S_0 = 50$, $n = 50$, $M = 200,000$).

r	σ	T	K	$\hat{C}^a$	CI ^a width	$\hat{C}^g$	CI ^g width	C_{BS}
0.00	0.20	2	35	15.0212	0.0716	14.6881	0.0703	15.5809
0.00	0.20	2	40	10.2577	0.0687	9.9882	0.0671	11.5413
0.00	0.20	2	45	6.1597	0.0599	5.9376	0.0580	8.2055
0.00	0.20	2	50	3.2404	0.0468	3.0682	0.0445	5.6231
0.00	0.20	2	55	1.4828	0.0325	1.3491	0.0301	3.7339
0.00	0.30	2	35	15.3100	0.1058	14.5939	0.1020	16.9581
0.00	0.30	2	40	11.0115	0.0996	10.4586	0.0948	13.5616
0.00	0.30	2	45	7.4803	0.0881	6.9675	0.0824	10.7207
0.00	0.30	2	50	4.8359	0.0747	4.4273	0.0688	8.3998
0.00	0.30	2	55	3.0127	0.0608	2.6586	0.0546	6.5379
0.00	0.40	2	35	15.8607	0.1406	14.7792	0.1309	18.7151
0.00	0.40	2	40	12.0877	0.1317	11.0699	0.1213	15.7672
0.00	0.40	2	45	8.8581	0.1195	7.9996	0.1079	13.2563
0.00	0.40	2	50	6.4325	0.1054	5.7317	0.0947	11.1351
0.00	0.40	2	55	4.5646	0.0921	3.9660	0.0809	9.3529
0.03	0.20	2	35	15.5901	0.0703	15.2443	0.0688	17.3933
0.03	0.20	2	40	11.0520	0.0680	10.7259	0.0662	13.3436
0.03	0.20	2	45	6.9664	0.0612	6.6883	0.0593	9.8601
0.03	0.20	2	50	3.9099	0.0501	3.6819	0.0474	7.0368
0.03	0.20	2	55	1.9254	0.0365	1.7532	0.0339	4.8699
0.04	0.40	2	35	16.3389	0.1381	15.2213	0.1280	20.4957
0.04	0.40	2	40	12.6411	0.1309	11.6265	0.1193	17.5260
0.04	0.40	2	45	9.5540	0.1203	8.6668	0.1081	14.9492
0.04	0.40	2	50	7.1081	0.1082	6.2874	0.0961	12.7332
0.04	0.40	2	55	5.2065	0.0961	4.5069	0.0835	10.8393
0.05	0.30	2	35	16.0801	0.1029	15.4269	0.0987	19.5267
0.05	0.30	2	40	12.0314	0.0987	11.4338	0.0933	16.0965
0.05	0.30	2	45	8.5586	0.0902	7.9701	0.0836	13.1201
0.05	0.30	2	50	5.7712	0.0783	5.3155	0.0721	10.5969
0.05	0.30	2	55	3.7789	0.0657	3.3563	0.0589	8.4976
1.00	0.10	2	35	17.0117	0.0193	13.6280	0.0131	45.2633
1.00	0.10	2	40	16.3385	0.0193	12.9474	0.0131	44.5866
1.00	0.10	2	45	15.6592	0.0194	12.2743	0.0131	43.9099
1.00	0.10	2	50	14.9763	0.0194	11.6002	0.0131	43.2332
1.00	0.10	2	55	14.3121	0.0194	10.9174	0.0131	42.5566

2.4 Question 3 — Confidence Intervals

Theorem 2.10 (95% confidence interval). *Let $s_M^2 = \frac{1}{M} \sum_{j=1}^M |X^{(j)} - \bar{X}_M|^2$. By the Central Limit Theorem,*

$$\frac{\bar{X}_M - \mathbb{E}[X]}{s_M/\sqrt{M}} \xrightarrow{\mathcal{L}} \mathcal{N}(0, 1).$$

An asymptotic 95% confidence interval for $\mathbb{E}[X]$ is

$$\text{CI}_{0.95} = \left[\bar{X}_M - 1.96 \frac{s_M}{\sqrt{M}}, \bar{X}_M + 1.96 \frac{s_M}{\sqrt{M}} \right], \quad |\text{CI}_{0.95}| = \frac{2 \times 1.96 s_M}{\sqrt{M}}. \quad (10)$$

Property 2.11 (Geometric mean reduces variance). By the AM-GM inequality, $A_T^{\text{geom}} \leq A_T^{\text{arith}}$ path-wise, hence

$$\text{Var}^{\mathbb{Q}}[(A_T^{\text{geom}} - K)^+] \leq \text{Var}^{\mathbb{Q}}[(A_T^{\text{arith}} - K)^+].$$

Consequently, the 95% CI for $\text{CallAsian}^{\text{geom}}$ is strictly shorter than for $\text{CallAsian}^{\text{arith}}$.

Figure 2 illustrates this variance reduction across all strikes. Numerically, for $r = 0.05$, $\sigma = 0.30$, $T = 1$, $M = 200,000$ (see Table 3), the 95% CI widths are:

- $K = 35$: arithmetic 0.0747, geometric 0.0730 (reduction $\approx 2.3\%$).
- $K = 40$: arithmetic 0.0720, geometric 0.0697 (reduction $\approx 3.2\%$).
- $K = 45$: arithmetic 0.0647, geometric 0.0621 (reduction $\approx 4.0\%$).
- $K = 50$: arithmetic 0.0527, geometric 0.0497 (reduction $\approx 5.7\%$).
- $K = 55$: arithmetic 0.0390, geometric 0.0358 (reduction $\approx 8.2\%$).

The reduction is larger for out-of-the-money strikes, where the geometric averaging effect is more pronounced. For longer maturity ($T = 2$) and higher volatility ($\sigma = 0.40$), the CI width reduction exceeds 10% (e.g. arithmetic 0.1406 vs. geometric 0.1309 for $r = 0$, $\sigma = 0.4$, $K = 35$, $T = 2$), consistent with Theorem 2.10.

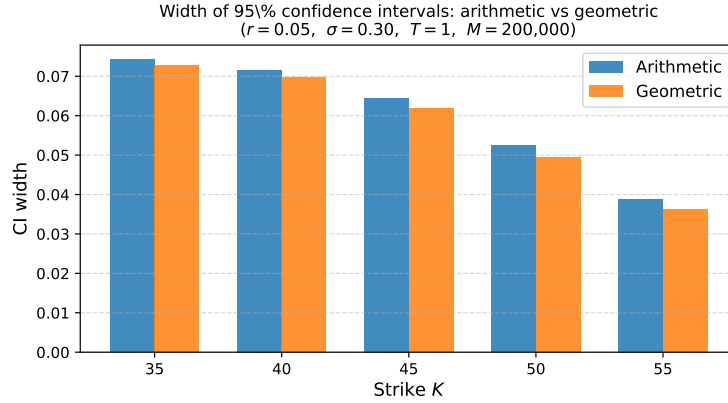


Figure 2: Length of 95% confidence intervals for arithmetic and geometric Asian calls as a function of K ($r = 0.05$, $\sigma = 0.30$, $T = 1$, $M = 200,000$). The geometric estimator consistently produces shorter intervals.

2.5 Question 4 — Comparison with the European Call

Lemma 2.12 (AM-GM inequality for averages). *For any non-negative real numbers $s_0, s_1, \dots, s_n > 0$,*

$$\left(\prod_{i=0}^n s_i \right)^{\frac{1}{n+1}} \leq \frac{1}{n+1} \sum_{i=0}^n s_i. \quad (11)$$

In particular, applied path-wise to the discretised asset prices, $A_T^{\text{geom}} \leq A_T^{\text{arith}}$ almost surely.

Proof. The AM-GM inequality follows from the concavity of $\ln$: by Jensen's inequality applied to the concave function $x \mapsto \ln x$,

$$\frac{1}{n+1} \sum_{i=0}^n \ln s_i \leq \ln \left(\frac{1}{n+1} \sum_{i=0}^n s_i \right).$$

Exponentiating both sides gives the result. □

Theorem 2.13 (Price ordering). *In the Black-Scholes model,*

$$\text{CallAsian}^{\text{geom}} \leq \text{CallAsian}^{\text{arith}} \leq \text{CallBS}. \quad (12)$$

Proof. The left inequality follows directly from Lemma 2.12: since $A_T^{\text{geom}} \leq A_T^{\text{arith}}$ almost surely and $(x - K)^+$ is non-decreasing, taking expectations under $\mathbb{Q}$ gives $\text{CallAsian}^{\text{geom}} \leq \text{CallAsian}^{\text{arith}}$.

For the right inequality, note that $(x - K)^+$ is convex in x . Since $A_T^{\text{arith}} = \frac{1}{T} \int_0^T S_t dt$, Jensen's inequality applied to the convex function $x \mapsto (x - K)^+$ gives

$$\left(\frac{1}{T} \int_0^T S_t dt - K \right)^+ \leq \frac{1}{T} \int_0^T (S_t - K)^+ dt.$$

Taking expectations under $\mathbb{Q}$ and discounting,

$$\text{CallAsian}^{\text{arith}} = e^{-rT} \mathbb{E}^{\mathbb{Q}} \left[\left(A_T^{\text{arith}} - K \right)^+ \right] \leq \frac{e^{-rT}}{T} \int_0^T \mathbb{E}^{\mathbb{Q}} [(S_t - K)^+] dt.$$

Since $t \mapsto e^{-rt} \mathbb{E}^{\mathbb{Q}}[(S_t - K)^+]$ is the price of a European call with maturity t , and the Black-Scholes call price is increasing in time to maturity (positive theta), we have $\mathbb{E}^{\mathbb{Q}}[(S_t - K)^+] \leq e^{r(T-t)} \mathbb{E}^{\mathbb{Q}}[(S_T - K)^+]$ for all $t \leq T$. Therefore,

$$\frac{e^{-rT}}{T} \int_0^T \mathbb{E}^{\mathbb{Q}} [(S_t - K)^+] dt \leq e^{-rT} \mathbb{E}^{\mathbb{Q}} [(S_T - K)^+] = \text{CallBS},$$

which completes the proof. $\square$

Remark 2.14. Intuitively, averaging smooths out extreme values of S , reducing the volatility of the payoff and hence its price relative to the European call.

The ordering is confirmed numerically in Tables 3 and 4, and illustrated in Figure 3. Several key observations emerge from the full results:

- (i) **Price ordering.** The inequality $C_{\text{Asian}}^g \leq C_{\text{Asian}}^a \leq C_{\text{BS}}$ holds for all 100 parameter combinations tested. For example, at $r = 0.05$, $\sigma = 0.30$, $T = 1$, $K = 50$: $C^g = 3.722 < C^a = 3.957 < C_{\text{BS}} = 7.116$.
- (ii) **Volatility effect.** All prices increase with σ . The gap between Asian and European prices widens: at $K = 40$, $T = 1$, $r = 0$, the difference $C_{\text{BS}} - C^a$ grows from 0.561 ($\sigma = 0.20$) to 2.344 ($\sigma = 0.40$), as averaging smooths out more volatility.
- (iii) **Maturity effect.** Longer maturities yield higher prices. At $r = 0.05$, $\sigma = 0.30$, $K = 40$: C^a increases from 10.961 ($T = 1$) to 12.031 ($T = 2$), and C_{BS} from 13.231 to 16.097.
- (iv) **High interest rate.** At $r = 1.00$, the European call substantially exceeds the Asian calls (e.g. $C_{\text{BS}} = 35.285$ vs. $C^a = 16.938$ at $K = 40$, $T = 1$), reflecting the large discounting effect and the fact that the average is taken over the entire path.

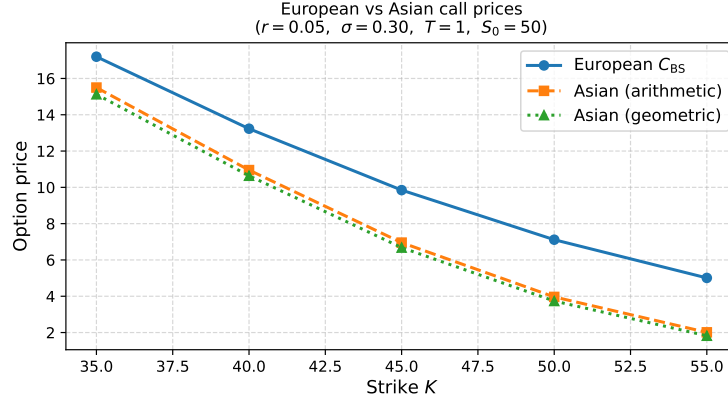


Figure 3: European vs. Asian call prices as functions of K ($r = 0.05$, $\sigma = 0.30$, $T = 1$, $S_0 = 50$). The inequality $\text{Geom} \leq \text{Arith} \leq \text{BS}$ is verified for all strikes.

3 Specific Topic: Optimal Market-Making under Fad Price Dynamics

This section is based on [3].

3.1 Introduction

3.1.1 Problem Overview

This section examines an optimal market-making problem where asset prices may temporarily deviate from their fundamental value because of short-term “fad” effects. In this setting, a market maker continuously posts bid and ask quotes, earns the bid–ask spread when orders arrive, and manages the risk associated with holding inventory. The main goal is to determine optimal quote displacements that maximise expected terminal wealth, while still penalising inventory risk.

Unlike the standard Avellaneda–Stoikov setup, this model includes both uninformed and informed traders. Uninformed traders generate liquidity demand in a random way, while informed traders respond to temporary price deviations and create adverse-selection risk. As a result, the market maker’s optimal quotes depend not only on current inventory, but also on the fad signal and, importantly, on whether that signal is observable.

3.1.2 Literature Review

This work follows the classical stochastic-control approach to market making [1]. In this framework, a dealer posts bid and ask quotes, takes on inventory risk from a diffusive mid-price, and receives market orders through Poisson arrival processes. Later studies extend this setup by adding inventory constraints and showing that the resulting Hamilton–Jacobi–Bellman (HJB) system can be converted into more tractable ordinary differential equations [7].

The paper also relates to the textbook by Cartea, Jaimungal and Penalva [5], which covers market making, optimal execution and informed trading using the same stochastic-control tools. Compared with those classical models, the key difference here is the mean-reverting fad component, which enters both the price SDE and the order arrival rates. This is what generates the distinction between informed and uninformed traders and creates the adverse-selection problem the market maker faces.

3.1.3 Main Challenges and Need for Stochastic Calculus

Stochastic calculus is needed here because the value function depends on two types of randomness at the same time: continuous diffusion dynamics from the fad U_t , and discrete jumps from the Poisson order arrivals N_t^a and N_t^b . The optimal control problem is handled through an HJB equation. In the partial-information case, the fad cannot be observed directly, so filtering theory is used to replace it with its conditional expectation given the observed price.

3.1.4 Techniques and Main Results

The problem is formulated as a stochastic control problem and analysed using the HJB framework. Because the exact HJB equation involves nonlinear jump terms, inventory constraints, and clipping terms, closed-form solutions are not available in general. To address this, the value function is characterised as a viscosity solution, and a tractable approximate full-information solution is developed.

For tractability, the approximation removes binding constraints and applies Taylor expansions to the exponential terms, yielding a quadratic ansatz for the value function:

$$V(t, q, u) = q^2 A(t) + qB(t, u) + C(t, u).$$

This transformation reduces the HJB equation to a Riccati equation for $A(t)$, while the remaining coefficients B and C satisfy linear ODEs.

In the partial-information setting, the fad is not directly observable, so it is replaced by its filtered estimate $\hat{U}_t$. The overall structure of the optimal quotes stays the same, but the signal variable is now replaced by this estimate. The main qualitative finding is that, in this model, adverse selection from informed traders is channelled entirely through quote skew (the asymmetry $\delta^{a,*} - \delta^{b,*}$), while the bid–ask spread $\delta^{a,*} + \delta^{b,*}$ remains invariant in the share of informed traders, unlike the classical Glosten–Milgrom intuition. Better information about the fad sharpens the skew response and raises the mean objective, with the full-information strategy outperforming both the partial-information and Cartea–Jaimungal–Penalva benchmarks at our parameters.

3.2 Mathematical and Model Setup

3.2.1 Probability Framework and Notation

Fix a finite trading horizon $\mathcal{T} = [0, T]$ with $T > 0$. We work on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \in \mathcal{T}}, \mathbb{P})$ that supports four mutually independent objects:

- two one-dimensional standard Brownian motions $(B_t)_{t \in \mathcal{T}}$ and $(Z_t)_{t \in \mathcal{T}}$;
- two unit-rate Poisson processes $(\bar{N}_t^a)_{t \in \mathcal{T}}$ and $(\bar{N}_t^b)_{t \in \mathcal{T}}$ (under a reference measure $\mathbb{P}_0$).

We take $(\mathcal{F}_t)_{t \in \mathcal{T}}$ to be the right-continuous, completed filtration jointly generated by these four processes. The controlled point processes $(N_t^a)_{t \in \mathcal{T}}$ and $(N_t^b)_{t \in \mathcal{T}}$ that appear throughout this section are obtained from $(\bar{N}^{a,b})$ via an absolutely continuous change of measure (a Doléans–Dade exponential), as in [3]; under the resulting measure $\mathbb{P}^\delta$, they become point processes whose stochastic intensities depend on the state in the way specified below. Expectations are taken under $\mathbb{P}^\delta$ unless we say otherwise, and we write this simply as $\mathbb{E}$.

Notation. Throughout, $t \in \mathcal{T}$ is the time index. A hat (e.g. $\hat{U}_t$) means the conditional expectation given a sub-filtration, and a, b label the ask and bid sides of the order book.

3.2.2 Price and Fad Dynamics

The mid-price $(S_t)_{t \in \mathcal{T}}$ and the latent fad $(U_t)_{t \in \mathcal{T}}$ are both exogenous diffusions, taken from [3]. The mid-price follows

$$dS_t = \mu dt + \sigma dW_t, \quad S_0 \in \mathbb{R}_+, \quad (13)$$

where $\mu \in \mathbb{R}$ and $\sigma > 0$. The driving Brownian motion is split into two pieces,

$$W_t = p Z_t + q U_t, \quad p, q > 0, \quad p^2 + q^2 = 1. \quad (14)$$

so that (Z_t) carries the classical fundamental information flow and (U_t) stands for the behaviour-driven, transient component of the price. The latter is itself a mean-reverting Ornstein–Uhlenbeck (OU) process,

$$dU_t = -\eta U_t dt + dB_t, \quad U_0 = 0, \quad (15)$$

with mean-reversion speed $\eta > 0$. The weight $q \in (0, 1)$ tunes how much of the price variability is due to the fad: a larger q means a noisier mid-price relative to its fundamental value.

Remark 3.1 (Fundamental price). Define the fundamental price $F_t := S_t - \sigma q U_t$. Substituting (14)–(15) into (13) and using $dU_t = -\eta U_t dt + dB_t$, one obtains the alternative representation

$$dS_t = (\mu - \eta \sigma q U_t) dt + \sigma d\bar{W}_t, \quad \bar{W}_t = p Z_t + q B_t,$$

where $\bar{W}$ is a Brownian motion correlated with B through $\langle \bar{W}, B \rangle_t = qt$.

3.2.3 Market Maker State Dynamics

The market maker posts unit-size limit orders on both sides of the book at all times. Her two controls are the bid and ask displacements $\delta_t^a, \delta_t^b \geq -\delta_\infty$, and from these she quotes the prices

$$S_t^a = S_t + \delta_t^a \quad (\text{ask}), \quad S_t^b = S_t - \delta_t^b \quad (\text{bid}). \quad (16)$$

A filled bid grows her inventory by one unit; a filled ask shrinks it by one. So the inventory and cash processes evolve as

$$dQ_t = dN_t^b - dN_t^a, \quad Q_0 \in \mathcal{Q}, \quad (17)$$

$$dX_t = (S_t + \delta_t^a) dN_t^a - (S_t - \delta_t^b) dN_t^b, \quad X_0 = 0. \quad (18)$$

Inventory cannot grow without bound: we impose hard limits $\underline{q} < 0 < \bar{q}$ and require $Q_t \in \mathcal{Q} = \{\underline{q}, \dots, \bar{q}\} \subset \mathbb{Z}$. Whenever a fill would push inventory past the limit on one side, the market maker pulls her quote on that side rather than trade, a constraint encoded by the indicators in the intensity formulas below.

3.2.4 Order Arrival Intensities

Order flow comes from two distinct populations of traders: uninformed traders, who only observe the mid-price, and informed traders, who can read the fad off the price and trade against the fundamental. The market maker sees only the aggregate, anonymous flow and cannot tell the two apart at the level of individual trades.

The stochastic intensities of the controlled counting processes (N_t^a) and (N_t^b) are

$$\lambda_t^a = \left(\underbrace{\phi e^{-k\delta_t^a}}_{\text{uninformed}} + \underbrace{\psi e^{-k\delta_t^a - \gamma(\sigma q U_t \vee S^-)}}_{\text{informed}} \right) \mathbf{1}_{\{Q_{t-} > \underline{q}\}}, \quad (19)$$

$$\lambda_t^b = \left(\underbrace{\phi e^{-k\delta_t^b}}_{\text{uninformed}} + \underbrace{\psi e^{-k\delta_t^b + \gamma(\sigma q U_t \wedge S^+)}}_{\text{informed}} \right) \mathbf{1}_{\{Q_{t-} < \bar{q}\}}, \quad (20)$$

where $k, \gamma > 0$, $\phi, \psi \geq 0$, and $S^- \leq 0 \leq S^+$ are the clipping levels. Capping the toxic term at $\sigma q U_t \vee S^-$ on the ask side (and similarly at $\sigma q U_t \wedge S^+$ on the bid) keeps the intensities bounded, which is what we need for the Doléans–Dade exponential in the construction of $\mathbb{P}^\delta$ to be a true martingale.

Economic interpretation. Uninformed traders only see the gap between the quote and the mid-price (δ_t^a on the ask, δ_t^b on the bid), so their arrival rate decays exponentially in the displacement at rate k . Informed traders, by contrast, can observe the fundamental $F_t = S_t - \sigma q U_t$ and react to the gap between the quote and that: from their point of view, the “generosity” of the ask is $S_t^a - F_t = \delta_t^a + \sigma q U_t$ and that of the bid is $F_t - S_t^b = \delta_t^b - \sigma q U_t$. A positive fad signals an over-priced asset and pulls in informed sellers; a negative one signals the opposite and brings in informed buyers. This is the channel through which adverse selection enters the market maker’s problem. The parameter γ controls how aggressively informed flow responds to the fad, while the ratio ψ/ϕ sets the relative size of the two populations.

3.2.5 Objective Function

The market maker is risk-averse in inventory and wants to maximise the performance criterion from [5],

$$J(\delta) = \mathbb{E} \left[X_T + Q_T S_T - \alpha Q_T^2 - \varphi \int_0^T Q_u^2 du \right], \quad (21)$$

over admissible controls $\delta = (\delta^a, \delta^b)$. Each term in (21) carries a clear economic meaning:

- X_T is the realised cash position at terminal time T ;
- $Q_T S_T$ is the mark-to-market value of any leftover inventory at the prevailing mid-price (we do not allow the market maker to recover the fundamental component when she liquidates);
- αQ_T^2 penalises non-zero terminal inventory; this is equivalent to a linear-in-size liquidation slippage, since $Q_T S_T - \alpha Q_T^2 = Q_T (S_T - \alpha Q_T)$;
- $\varphi \int_0^T Q_u^2 du$ is a running inventory penalty that pushes the market maker towards a flat book throughout the horizon.

The set of admissible controls is

$$\mathcal{U} := \{ \delta = (\delta^a, \delta^b) \in L^2(\Omega \times [0, T]) : \delta \geq -\delta_\infty, \mathcal{F}\text{-predictable} \}.$$

3.3 Optimisation Problem and Relevant Quantities

3.3.1 Controlled SDE System

Putting together (13), (15), (17)–(18) and the intensities (19)–(20), the full controlled state $\mathbf{X}_t = (X_t, Q_t, S_t, U_t)$ obeys

$$\begin{cases} dX_t = (S_t + \delta_t^a) dN_t^a - (S_t - \delta_t^b) dN_t^b, & X_0 = 0, \\ dQ_t = dN_t^b - dN_t^a, & Q_0 \in \mathcal{Q}, \\ dS_t = \mu dt + \sigma(p dZ_t + q dU_t), & S_0 \in \mathbb{R}_+, \\ dU_t = -\eta U_t dt + dB_t, & U_0 = 0. \end{cases} \quad (22)$$

Note that we do not include Z_t in the state, since it is recoverable from (S_t, U_t) via $Z_t = \sigma^{-1} p^{-1} (S_t - S_0 - \mu t - \sigma q U_t)$. Two of the components (S and U) are diffusions and two (N^a , N^b) are controlled point processes, so the Itô formula we will need below has both Brownian and jump contributions.

3.3.2 The Stochastic Control Problem

The market maker’s task is to pick $\delta \in \mathcal{U}$ that maximises the criterion (21). Defining the value function

$$V(t, q, u) := \sup_{\delta \in \mathcal{U}} \mathbb{E}_{t, q, u} \left[-\alpha Q_T^2 + \int_t^T (\delta_s^a \lambda_s^a + \delta_s^b \lambda_s^b - \varphi Q_s^2 + Q_s \mu - \eta q \sigma Q_s U_s) ds \right], \quad (23)$$

the original problem $\sup_{\delta} J(\delta)$ becomes equivalent to evaluating $V(0, Q_0, 0)$, after the Itô reduction we carry out in Section 3.4. We use $\mathbb{E}_{t,q,u}$ as shorthand for the conditional expectation given $Q_t = q$ and $U_t = u$.

3.3.3 Relevant Quantities of Interest

The main things we look at are:

- the optimal feedback controls $\delta^{a,*}(t, q, u)$ and $\delta^{b,*}(t, q, u)$, written as finite differences of V in q ;
- the value function $V(t, q, u)$ itself, which under the quadratic ansatz of Section 3.4 collapses to three deterministic functions $A(t)$, $B(t, u)$, and $C(t, u)$;
- the equilibrium bid–ask spread $\delta^{a,*} + \delta^{b,*}$ and how it varies with the model parameters $(q, \gamma, \eta, \phi, \psi)$;
- the inventory and cash trajectories that the optimal policy produces, together with the resulting profit-and-loss distribution.

3.3.4 Information Structure

One important feature of this model is that the same problem can be run under different assumptions about what the market maker can observe. We compare three cases following [3]:

- (i) **Full information (FI).** The market maker observes the entire filtration $\mathcal{F} = (\mathcal{F}_t)_{t \in \mathcal{T}}$, and in particular the realised fad path (U_t) . Her optimal control is a feedback function of (t, Q_{t-}, U_t) .
- (ii) **Partial information (PI).** The market maker only sees the price filtration $\mathcal{F}^S = (\mathcal{F}_t^S)_{t \in \mathcal{T}}$ generated by (S_t) (and the order flow). The fad (U_t) is hidden, and the best she can do is replace it by its conditional expectation $\hat{U}_t = \mathbb{E}[U_t \mid \mathcal{F}_t^S]$, computed by a Kalman–Bucy filter. Her optimal control is then a feedback function of $(t, Q_{t-}, \hat{U}_t)$.
- (iii) **CJP benchmark.** The market maker uses the Cartea–Jaimungal–Penalva strategy [5, Chapter 10], which ignores the fad entirely (it assumes $q = 0$ and $\gamma = 0$). Her control depends only on (t, Q_{t-}) . The actual market does have a fad, so the CJP strategy is computed under a misspecified model and serves as a natural baseline.

Comparing these three strategies in Section 3.5 tells us about the value of information: the gap between FI and PI tells us what the market maker loses by not seeing the fad directly, and the gap between PI and CJP tells us how much it pays to model the fad correctly even when it cannot be observed.

3.4 Solutions

Following [3], maximising $J(\delta)$ is equivalent to maximising the value function $V(t, q, u)$ defined in (23), up to the constant $Q_0 S_0$. We work with V directly.

3.4.1 Dynamic Programming and the HJB Equation

By the standard dynamic programming principle for jump-diffusion control problems [9, Ch. 5], the value function V formally satisfies a Hamilton–Jacobi–Bellman (HJB) equation. The infinitesimal generator of the state splits naturally into two pieces: a diffusion part inherited from U , and a jump part inherited from N^a and N^b .

Theorem 3.2 (HJB equation). *The value function $V(t, q, u)$ defined in (23) formally solves the HJB equation*

$$\begin{aligned} 0 = & \partial_t V - \eta u \partial_u V + \frac{1}{2} \partial_{uu}^2 V - \varphi q^2 + q(\mu - \eta \sigma q u) \\ & + \sup_{\delta^a} \left\{ e^{-k\delta^a} (\phi + \psi e^{-\gamma(\sigma q u \vee S^-)}) [\delta^a + V(t, q-1, u) - V(t, q, u)] \right\} \mathbf{1}_{\{q > \underline{q}\}} \\ & + \sup_{\delta^b} \left\{ e^{-k\delta^b} (\phi + \psi e^{\gamma(\sigma q u \wedge S^+)}) [\delta^b + V(t, q+1, u) - V(t, q, u)] \right\} \mathbf{1}_{\{q < \bar{q}\}}, \end{aligned} \quad (24)$$

with terminal condition $V(T, q, u) = -\alpha q^2$.

The four lines of (24) read as follows. The first line collects the deterministic drift and the running inventory penalty; the second line is the OU diffusion generator acting in u ; and the last two “sup” terms are the jump contributions on the ask and bid sides, where the inventory q jumps by ± 1 each time an order is filled.

3.4.2 Optimal Quote Displacements

Each supremum in (24) has the same generic form, $\max_{\delta} e^{-k\delta} (\delta + V_*)$, where V_* is a finite difference of the value function in q . The first-order condition gives

$$\frac{d}{d\delta} [e^{-k\delta} (\delta + V_*)] = e^{-k\delta} [1 - k(\delta + V_*)] = 0 \implies \delta = \frac{1}{k} - V_*.$$

Proposition 3.3 (Optimal feedback controls). *Whenever the lower bound $-\delta_\infty$ is non-binding, the unique maximisers of (24) are*

$$\delta^{a,*}(t, q, u) = \frac{1}{k} - [V(t, q-1, u) - V(t, q, u)], \quad q \neq \underline{q}, \quad (25)$$

$$\delta^{b,*}(t, q, u) = \frac{1}{k} - [V(t, q+1, u) - V(t, q, u)], \quad q \neq \bar{q}. \quad (26)$$

The constraint is enforced by capping at $-\delta_\infty$ if necessary.

The optimal displacement on each side is the elasticity term $1/k$, minus the marginal value of holding one extra unit of inventory in the appropriate direction.

3.4.3 Viscosity Solution Characterisation

The HJB equation (24) is a non-local, non-smooth PDE: the jump terms link V at (t, q, u) to V at $(t, q \pm 1, u)$, and the inventory-bound indicators bring discontinuities into q . As a result, V cannot generally be a classical $C^{1,2}$ solution, and we have to work with the weaker notion of a viscosity solution [10, Ch. 5].

Let $V_*(t, q, u) = \liminf V$ and $V^*(t, q, u) = \limsup V$ denote the lower and upper semicontinuous envelopes (limits taken at (t, q, u)), and write the Hamiltonian as

$$\begin{aligned} H(q, u, V_1, V_2) := & \sup_{\delta^a} (\phi + \psi e^{-\gamma(\sigma q u \vee S^-)}) e^{-k\delta^a} (\delta^a + V_1) \mathbf{1}_{\{q > \underline{q}\}} \\ & + \sup_{\delta^b} (\phi + \psi e^{\gamma(\sigma q u \wedge S^+)}) e^{-k\delta^b} (\delta^b + V_2) \mathbf{1}_{\{q < \bar{q}\}}. \end{aligned} \quad (27)$$

Theorem 3.4 (Viscosity characterisation). *The value function V is locally bounded on $E := [0, T] \times \mathcal{Q} \times \mathbb{R}$, and is a viscosity solution of (24) in the sense that:*

- (Sub-solution.) For every $\Phi \in C^{1,2}$ such that $V^* - \Phi$ has a local maximum at (t_0, q_0, u_0) with value zero, $F(t_0, q_0, u_0, \Phi, \partial_t \Phi, \partial_u \Phi, \partial_{uu}^2 \Phi) \geq 0$.
- (Super-solution.) For every $\Phi \in C^{1,2}$ such that $V_* - \Phi$ has a local minimum at (t_0, q_0, u_0) with value zero, $F(t_0, q_0, u_0, \Phi, \partial_t \Phi, \partial_u \Phi, \partial_{uu}^2 \Phi) \leq 0$.

Here F is the functional given by the right-hand side of (24) written compactly using H .

3.4.4 Approximate Full-Information Solution

There is no closed-form solution for the HJB equation (24), so we follow [3, 4] and develop a tractable approximation. The plan is to:

- (i) drop the inventory bounds ($\underline{q} = -\infty, \bar{q} = +\infty$) and the lower bound on displacements ($\delta_\infty = \infty$);
- (ii) drop the clipping ($S^- = -\infty, S^+ = +\infty$);
- (iii) Taylor-expand $p \mapsto k^{-1}e^{kp}$ to second order around zero, applied to the finite differences $\Delta^\pm V := V(t, q \pm 1, u) - V(t, q, u)$;
- (iv) Taylor-expand $u \mapsto e^{\pm\gamma\sigma qu}$ to first order around zero (valid for $\gamma\sigma qu$ small).

After these four reductions the HJB equation becomes a smooth polynomial PDE that admits a quadratic-in- q ansatz.

Proposition 3.5 (Quadratic ansatz). *Suppose $V(t, q, u) = q^2 A(t) + qB(t, u) + C(t, u)$, where $A : [0, T] \rightarrow \mathbb{R}$ and $B, C : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$ solve the system*

$$\begin{cases} \partial_t A - \varphi + 4(\phi + \psi)e^{-1}k A^2 = 0, \\ \partial_t B + \frac{1}{2}\partial_{uu}^2 B + (\mu - \eta\sigma qu) - \eta u \partial_u B + 4(\phi + \psi)e^{-1}k A B \\ \quad + 4e^{-1}\psi q\sigma\gamma u A + 4e^{-1}k q\gamma\sigma u \psi A^2 = 0, \\ \partial_t C + \frac{1}{2}\partial_{uu}^2 C - \eta u \partial_u C + e^{-1}k^{-1}(2(\phi + \psi) + 2kA(\phi + \psi) + 2\psi k B q\sigma\gamma u \\ \quad + k^2(\phi + \psi)(A^2 + B^2) + 2k^2\psi q\sigma\gamma u AB) = 0, \end{cases} \quad (28)$$

with terminal conditions $A(T) = -\alpha$, $B(T, u) = 0$, $C(T, u) = 0$. Then V solves the approximate HJB equation.

The first line of (28) is a scalar Riccati ODE for $A(t)$. Setting $\kappa := 4(\phi + \psi)e^{-1}k$, its closed-form solution (in the generic case $\sqrt{\varphi} \neq \sqrt{\kappa}\alpha$) is

$$A(t) = \sqrt{\varphi/\kappa} \frac{1 - e^{2\sqrt{\varphi\kappa}(T-t)} \beta}{1 + e^{2\sqrt{\varphi\kappa}(T-t)} \beta}, \quad \beta = \frac{\sqrt{\varphi} + \sqrt{\kappa}\alpha}{\sqrt{\varphi} - \sqrt{\kappa}\alpha}. \quad (29)$$

Proposition 3.6 (Reduction to a linear ODE system). *Postulate $B(t, u) = b_0(t) + u b_1(t)$ and $C(t, u) = c_0(t) + u c_1(t) + u^2 c_2(t)$. Then $(b_0, b_1, c_0, c_1, c_2)$ solve a linear ODE system with terminal condition zero. The system can be solved by the Duhamel formula and straightforward numerical integration.*

Plugging the quadratic ansatz back into (25)–(26) gives the optimal feedback controls in compact form:

$$\delta^{a,*}(t, q, u) = \frac{1}{k} + (2q - 1)A(t) + B(t, u), \quad \delta^{b,*}(t, q, u) = \frac{1}{k} - (2q + 1)A(t) - B(t, u). \quad (30)$$

Two consequences are immediate:

- the optimal spread, $\delta^{a,*} + \delta^{b,*} = 2/k - 2A(t)$, depends only on $A(t)$ and therefore only on the structural parameters $(\varphi, \phi, \psi, k, \alpha)$, never on the fad u . This is what explains the observation in Figure 8b of [3] that the spread is invariant in the share of informed traders;
- the asymmetry (or skew) between the two quotes is $\delta^{a,*} - \delta^{b,*} = 2(qA(t) + B(t, u))$, which carries all of the dependence on inventory q and on the fad signal u . So in the FI strategy, inventory and signal information enter only through the skew, never through the spread.

3.4.5 Partial-Information Solution via Filtering

In the partial-information regime the market maker no longer observes (U_t) directly. She has to infer it from the price filtration $\mathcal{F}^S$. We follow the classical Kalman–Bucy paradigm for linear-Gaussian filtering [2], and define the conditional mean and conditional variance

$$\hat{U}_t := \mathbb{E}[U_t | \mathcal{F}_t^S], \quad \hat{P}_t := \mathbb{E}[(U_t - \hat{U}_t)^2 | \mathcal{F}_t^S]. \quad (31)$$

Define the innovation process

$$dI_t = dS_t - \mathbb{E}[\mu - \eta\sigma q U_t | \mathcal{F}_t^S] dt,$$

which, after rescaling by σ^{-1} , is an $\mathcal{F}^S$ -Brownian motion [2].

Theorem 3.7 (Kalman–Bucy filter for the fad). *The conditional mean $\hat{U}_t$ satisfies the SDE*

$$d\hat{U}_t = -\eta\hat{U}_t dt + \sigma^{-1}(-\eta q \hat{P}_t + q) dI_t, \quad \hat{U}_0 = 0. \quad (32)$$

The conditional variance $\hat{P}_t$ is deterministic and solves the Riccati ODE

$$\frac{d}{dt}\hat{P}_t = -\eta^2 q^2 \hat{P}_t^2 - \hat{P}_t(2\eta - 2\eta q^2) + p^2, \quad \hat{P}_0 = 0. \quad (33)$$

Once we have the filter, the partial-information control problem is formally identical to the full-information one, with U_t replaced by $\hat{U}_t$ throughout. The market maker now solves

$$V^{\text{PI}}(t, q, \hat{u}) = \sup_{\delta \in \mathcal{U}^S} \mathbb{E}_{t, q, \hat{u}} \left[-\alpha Q_T^2 + \int_t^T (\delta_s^a \lambda_s^a + \delta_s^b \lambda_s^b - \varphi Q_s^2 + (\mu - \eta q \sigma \hat{U}_s) Q_s) ds \right], \quad (34)$$

where $\mathcal{U}^S$ is the set of admissible controls predictable with respect to the enlarged filtration generated by $(\tilde{I}_t, \hat{U}_t, N_t^{a,b})$, and the intensities are now given by [3]

$$\begin{aligned} \lambda_t^a &= (\phi + \psi e^{-\gamma(\sigma q \hat{U}_t \vee S^-)}) e^{-k\delta_t^a} \mathbf{1}_{\{Q_{t-} > \bar{q}\}}, \\ \lambda_t^b &= (\phi + \psi e^{\gamma(\sigma q \hat{U}_t \wedge S^+)}) e^{-k\delta_t^b} \mathbf{1}_{\{Q_{t-} < \bar{q}\}}. \end{aligned}$$

The corresponding HJB equation has the same algebraic structure as (24), except that the second-order term in $\hat{u}$ now carries the diffusion coefficient $(\hat{P}_t q \eta - q)^2$. Since the volatility coefficients do not enter the linear or quadratic-in- q terms of the value function, the optimal feedback controls under partial information have exactly the same form as in (30), with u replaced by $\hat{u}$ and the ODE coefficients (A, B) adjusted accordingly:

$$\delta^{a,*\text{PI}}(t, q, \hat{u}) = \frac{1}{k} + (2q - 1)A(t) + B(t, \hat{u}), \quad \delta^{b,*\text{PI}}(t, q, \hat{u}) = \frac{1}{k} - (2q + 1)A(t) - B(t, \hat{u}). \quad (35)$$

In short, the separation principle says: estimate the hidden state first, then act as if that estimate were the true state. As [3] points out, this is not in general optimal for non-linear-quadratic problems. In our simulations the PI mean objective is $J_{\text{PI}} = 21.20$ at baseline ($q = 0.6$), against $J_{\text{FI}} = 21.33$ (Table 5), so the loss from substituting the filter for the unobserved U_t is 0.13 in absolute units, or about 0.6% of the FI objective. Figure 7 shows that this gap shrinks further as q increases and effectively vanishes at $q = 1$, where Proposition 22 of [3] gives $\hat{U}_t = U_t$ exactly.

3.5 Numerical Simulations

3.5.1 Strategy Comparison

We benchmark the three strategies introduced in Section 3.3 (FI, PI, CJP) on simulated paths of the full model. Baseline parameters follow [3]: $T = 1$, $S_0 = 100$, $X_0 = Q_0 = 0$, $\sigma = 1$, $\eta = 10$, $q = 0.6$ (so $p = \sqrt{1 - q^2} \approx 0.8$), $\mu = 0$, $k = \gamma = 1$, $\phi = \psi = 15$, $\alpha = 0.1$ and $\varphi = 0.1$, with inventory bounds $\underline{q} = -10$, $\bar{q} = 10$ and lower displacement bound $\delta_\infty = 5$. We discretise the horizon on a uniform grid of $N = 1000$ time steps and average our Monte Carlo estimates over $M = 3000$ independent paths. The coefficients $A(t)$, $b_0(t)$, $b_1(t)$ in (28) are pre-computed numerically on a fine auxiliary grid.

Performance criterion. For each path we record the realised value of the objective (21) together with the time-averaged spread $\bar{s} := T^{-1} \int_0^T (\delta_t^{a,*} + \delta_t^{b,*}) dt$. Means and standard deviations are reported in Table 5. The ordering matches the prediction of [3]: Full Information delivers the highest mean objective, Partial Information comes second, and the CJP benchmark trails:

$$J_{\text{FI}} \geq J_{\text{PI}} \geq J_{\text{CJP}}.$$

Standard deviations are nearly identical across the three strategies, which says inventory risk is comparable. The time-average spread is identical across all three policies, consistent with the analytic result $\delta^{a,*} + \delta^{b,*} = 2/k - 2A(t)$. So the performance differences come entirely through quote skew, never through spread adjustments.

Table 5: Average performance and average quoted spread for the three strategies (baseline parameters, $M = 3000$ paths). Standard deviations in parentheses.

Strategy	Mean objective	Mean spread
Full Information (FI)	21.33 (4.94)	2.0930
Partial Information (PI)	21.20 (4.95)	2.0930
CJP benchmark	21.18 (4.94)	2.0930

3.5.2 Graphs and Interpretation

We now walk through the eight diagnostic plots produced by our simulation code. Each plot isolates a single driver of the model.

Optimal quote displacements vs. fad. Figure 4 (the analogue of Figure 1 in [3]) plots $\delta^{a,*}$ and $\delta^{b,*}$ as linear functions of u , with one curve per inventory level q . Two features stand out. First, the ask is decreasing in u while the bid is increasing, which reflects the market maker’s effort to balance informed flow across the two sides. Second, the slope grows in $|q|$ and in q (the weight of the fad in the price). The symmetry of these linear shifts confirms the analytic statement that the spread $\delta^{a,*} + \delta^{b,*}$ is constant in u , while the skew $\delta^{a,*} - \delta^{b,*}$ moves with u .

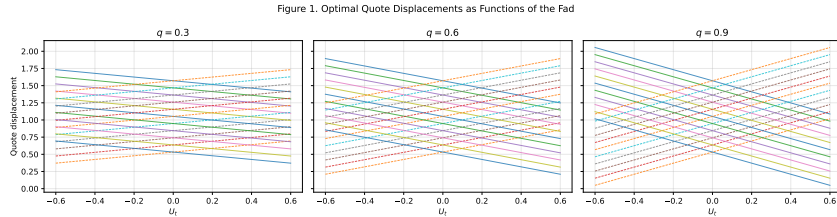


Figure 4: Optimal ask (solid) and bid (dashed) displacements as functions of the fad u , for several inventory levels q encoded by colour. The ask is decreasing in u and the bid is increasing, while their sum (the spread) is constant.

Inventory paths under different γ . Figure 5 shows one realisation of (U_t) and the corresponding (Q_t) at two values of γ : $\gamma = 0.1$ on the left and $\gamma = 10$ on the right. The fad path is the same in both panels (it does not depend on γ), but the inventory path under $\gamma = 10$ is visibly more volatile and far more tightly correlated with U . This is exactly what we would expect: when informed flow is more sensitive to mispricing, the market maker accumulates inventory more aggressively in the direction the fad dictates.

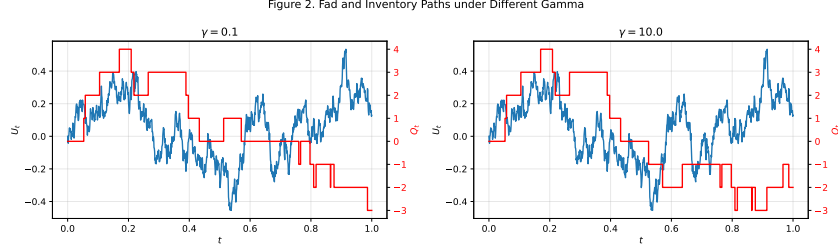


Figure 5: One realised path of (U_t, Q_t) under the FI strategy for $\gamma = 0.1$ (left) and $\gamma = 10$ (right). The fad is identical; the inventory under $\gamma = 10$ tracks the fad much more closely.

Filtered fad vs. true fad. Figure 6 shows one realisation of U_t alongside its Kalman–Bucy filter $\hat{U}_t$ from (32), for three values of $q \in \{0.3, 0.6, 0.9\}$. The filter is smoother than the truth and lags it slightly. As q grows, the price becomes a more informative signal of the fad and the filter tracks U_t more closely; in the limit $q \rightarrow 1$ the filter coincides with the truth (Proposition 22 of [3]), while at $q \rightarrow 0$ it collapses to its long-run mean of zero.



Figure 6: True fad U_t (blue) and Kalman–Bucy filter $\hat{U}_t$ (red) for $q = 0.3, 0.6, 0.9$. As $q \uparrow 1$ the filter becomes accurate; as $q \downarrow 0$ it collapses to zero.

Performance vs. q . Figure 7 plots the average objective against q for the three strategies. Three regimes emerge. For small q , the three strategies are essentially indistinguishable: the fad carries too little weight in the price and in the arrivals to matter. At moderate q , FI starts to enjoy a meaningful edge while PI and CJP stay close to each other. As $q \rightarrow 1$, FI and PI converge (the filter becomes exact) while CJP deteriorates, because the misspecification cost grows with q .

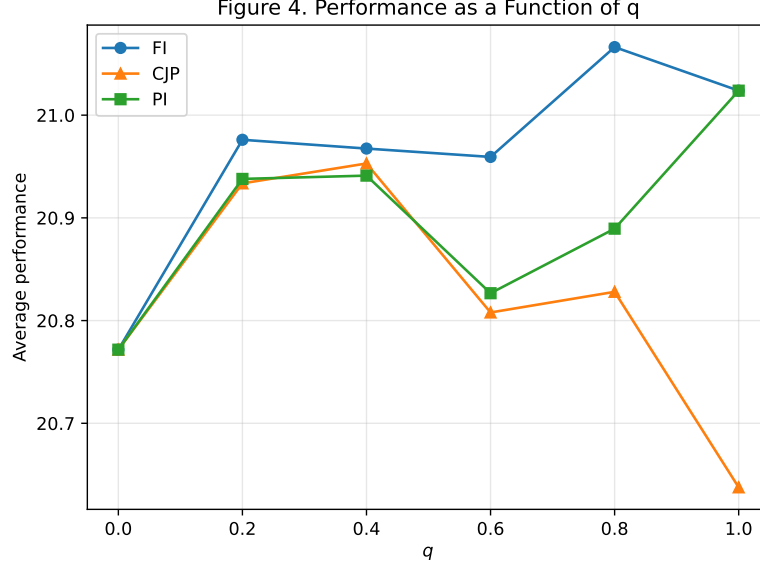


Figure 7: Average performance of the three strategies as q varies, with ψ rescaled to keep the expected number of arrivals constant. FI dominates uniformly; PI and FI agree at $q = 1$; CJP degrades fastest.

Performance vs. γ . Figure 8 plots the average objective against γ , with ψ rescaled so that the expected number of arrivals is held constant (cf. Eq. (61) of [3]). All three curves decrease quasi-linearly: as informed flow becomes more toxic, the market maker takes a hit regardless of how much she sees. The ranking $FI \geq PI \geq CJP$ is preserved throughout.

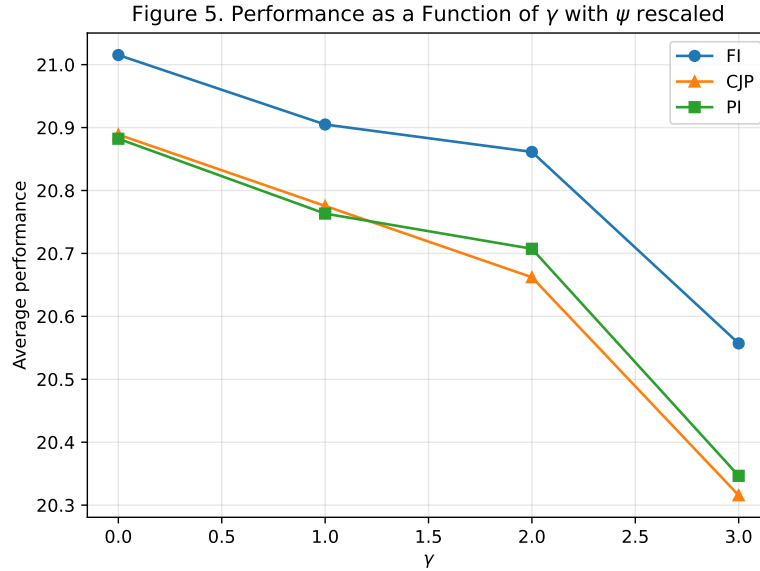


Figure 8: Average performance as a function of γ with ψ rescaled. Higher γ means sharper informed orders and lower performance for all three strategies.

Performance vs. η . Figure 9 plots the average objective against η . A larger η means the fad reverts to zero faster, the unconditional variance of U shrinks, and all three strategies improve. The gap between FI and CJP narrows in this regime, because for fast mean reversion the (misspecified) CJP model is itself a better approximation of the truth.

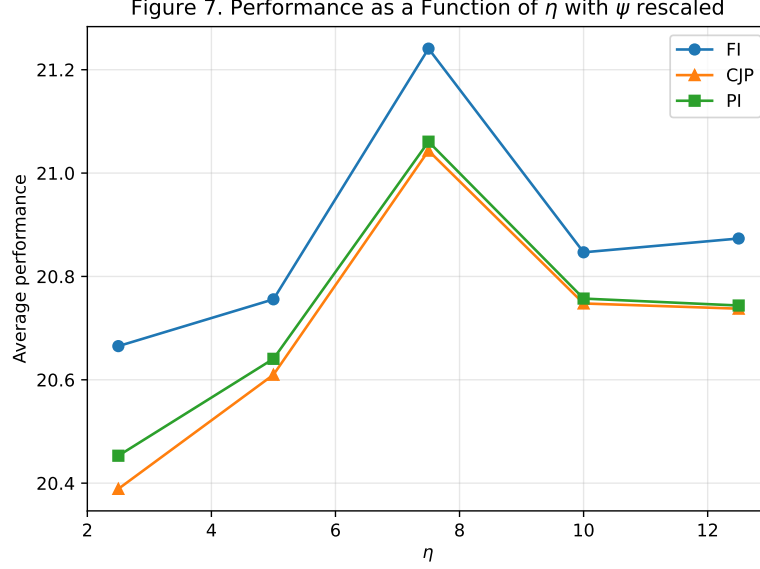


Figure 9: Average performance as a function of the mean-reversion speed η . Faster mean reversion shrinks the variance of the fad and improves all strategies; the gap between FI and CJP narrows.

Quote asymmetry vs. informed share. Figure 10 keeps the expected number of arrivals fixed at 30 and varies $\phi/(\phi + \psi)$, the share of uninformed flow, from 0% to 100%. The total spread $\delta^{a,*} + \delta^{b,*}$ is flat across all configurations (panel b), but the absolute skew $|\delta^{a,*} - \delta^{b,*}|$ at a fixed $u \neq 0$ grows monotonically with the informed share (panel c). Performance (panel a) drops as the informed share rises, in classical adverse-selection style à la [6]. The takeaway is clear: in this model, all adverse-selection effects pass through quote skew, never through spread widening.

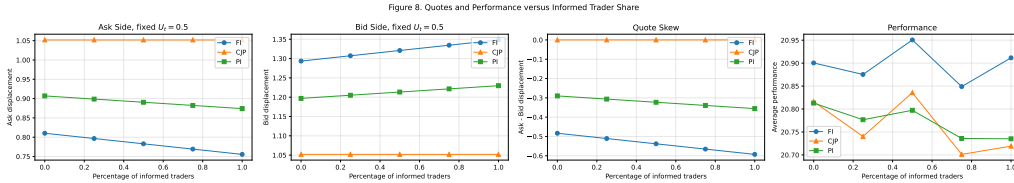


Figure 10: Performance (a), spread (b) and absolute skew (c) as functions of the share of informed traders. The spread is invariant; performance drops and skew rises with the informed share.

Distribution of terminal objective. Figure 11 shows kernel-density estimates of the terminal-objective distribution under each of the three strategies. All three densities are roughly Gaussian and overlap heavily; the FI density is shifted a little to the right, but the dispersion is essentially the same across strategies. Looking at the numbers, FI shifts the mean by 0.15 relative to CJP (21.33 vs 21.18) while the standard deviation stays at roughly 4.94 across all three. Better information raises average performance without changing the risk profile.

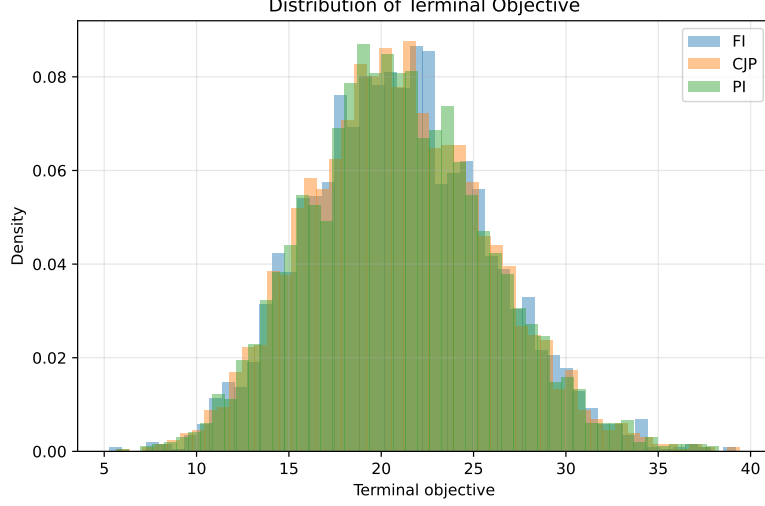


Figure 11: Kernel-density estimates of the realised objective under each of the three strategies. The densities are roughly Gaussian and overlap heavily; FI is slightly shifted to the right.

Bottom line. Three predictions of Theorems 3.2–3.7 and Propositions 3.3–3.6 are borne out by the simulations: (i) the time-averaged spread is identical to four decimals across FI, PI, and CJP ($\bar{s} = 2.0930$ in Table 5), matching the closed-form invariance $\delta^{a,*} + \delta^{b,*} = 2/k - 2A(t)$ from (30); (ii) the ordering $J_{\text{FI}} \geq J_{\text{PI}} \geq J_{\text{CJP}}$ holds in mean at baseline (21.33, 21.20, 21.18) and the FI–CJP gap widens with q in Figure 7, with the cost of misspecification growing fastest in the fad-dominated regime $q \rightarrow 1$; (iii) the Kalman–Bucy filter from Theorem 3.7 closes the FI–PI portion of that gap to within numerical noise as $q \rightarrow 1$, consistent with the $\hat{U}_t \rightarrow U_t$ convergence visible in Figure 6.

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A Python Code for Section 2

All numerical results and figures in Section 2 were produced using Python 3 with `numpy`, `scipy.stats`, and `matplotlib`. The random seed was fixed to 42 for reproducibility. Parameters used throughout: $S_0 = 50$, $n = 50$ time steps, $M = 200,000$ simulations.

A.1 Black-Scholes Call Price (Question 1)

Listing 1: Black-Scholes European call price and convexity check.

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 from scipy.stats import norm
4
5 np.random.seed(42)
6
7 def CallBS(S0, K, r, sigma, T):
8     """Black-Scholes European call option price."""
9     d1 = (np.log(S0 / K) + (r + 0.5 * sigma**2) * T) / (sigma * np.sqrt(T))
10    d2 = d1 - sigma * np.sqrt(T)
11    return S0 * norm.cdf(d1) - K * np.exp(-r * T) * norm.cdf(d2)
12
13 # --- Convexity check ---
14 K_grid = np.linspace(35, 55, 200)
15 C_grid = np.array([CallBS(50, K, 0.05, 0.30, 1) for K in K_grid])
16 second_diff = C_grid[:-2] - 2 * C_grid[1:-1] + C_grid[2:]
17 print(f"Min second finite difference: {second_diff.min():.4e}") # > 0
18
19 # --- Plot ---
20 fig, ax = plt.subplots(figsize=(7, 4))
21 ax.plot(K_grid, C_grid, linewidth=2)
22 ax.set_xlabel("Strike $K$"); ax.set_ylabel("$C_{BS}(K)$")
23 ax.set_title("Black-Scholes call price vs $K$ "
24             f"$r=0.05$, $\sigma=0.30$, $T=1$")
25 ax.grid(True, linestyle="--", alpha=0.5)
26 plt.tight_layout()
27 plt.savefig("fig_q1_convexity.pdf", bbox_inches="tight")
```

A.2 GBM Path Simulation

Listing 2: Exact log-normal GBM path simulation under the risk-neutral measure.

```
1 def simulate_paths(S0, r, sigma, T, n_steps, M):
2     """
3     Simulate M GBM paths on a uniform grid of n_steps+1 dates.
4     Returns an (M, n_steps+1) array of asset prices.
5     """
6     dt = T / n_steps
7     Z = np.random.randn(M, n_steps) # i.i.d. N(0,1) increments
8     S = np.zeros((M, n_steps + 1))
9     S[:, 0] = S0
10    for t in range(1, n_steps + 1):
11        S[:, t] = S[:, t-1] * np.exp(
12            (r - 0.5 * sigma**2) * dt + sigma * np.sqrt(dt) * Z[:, t-1]
13        )
14    return S
```

A.3 Asian Option Monte Carlo Pricer (Questions 2 & 3)

Listing 3: Monte Carlo Asian call pricer with 95% confidence interval.

```

1 def asian_option_mc(S0, K, r, sigma, T,
2                     n_steps=50, M=200_000, method="arithmetic"):
3     """
4     Price an Asian call via Monte Carlo.
5
6     Parameters
7     -----
8     method : {"arithmetic", "geometric"}
9
10    Returns
11    -----
12    price : float    discounted MC estimate
13    std : float     sample std of discounted payoffs
14    """
15    S = simulate_paths(S0, r, sigma, T, n_steps, M)
16
17    if method == "arithmetic":
18        A = np.mean(S, axis=1)
19    elif method == "geometric":
20        A = np.exp(np.mean(np.log(S), axis=1))
21    else:
22        raise ValueError("method must be 'arithmetic' or 'geometric'")
23
24    disc_payoff = np.exp(-r * T) * np.maximum(A - K, 0)
25    return disc_payoff.mean(), disc_payoff.std(ddof=1)
26
27
28 def confidence_interval_95(mean, std, M):
29     """Return (lower, upper, width) of the 95% MC confidence interval."""
30     hw = 1.96 * std / np.sqrt(M)
31     return mean - hw, mean + hw, 2 * hw

```

A.4 Full Parameter Sweep

Listing 4: Parameter sweep over all required combinations (Tables 3–4).

```

1 S0, M, n_steps = 50, 200_000, 50
2
3 param_rows = [
4     (0.00, [0.2, 0.3, 0.4]),
5     (0.03, [0.2]),
6     (0.04, [0.4]),
7     (0.05, [0.3]),
8     (1.00, [0.1]),
9 ]
10 K_list = [35, 40, 45, 50, 55]
11 T_list = [1, 2]
12
13 rows = []
14 for r, sigma_list in param_rows:
15     for sigma in sigma_list:
16         for T in T_list:
17             for K in K_list:
18                 pa, sa = asian_option_mc(S0, K, r, sigma, T,
19                                         n_steps, M, "arithmetic")
20                 pg, sg = asian_option_mc(S0, K, r, sigma, T,
21                                         n_steps, M, "geometric")
22                 la, ua, len_a = confidence_interval_95(pa, sa, M)
23                 lg, ug, len_g = confidence_interval_95(pg, sg, M)
24                 rows.append(dict(
25                     r=r, sigma=sigma, T=T, K=K,

```

```

26         Arith=pa, CI_A_len=len_a,
27         Geom=pg, CI_G_len=len_g,
28         BS=CallBS(S0, K, r, sigma, T)
29     ))

```

A.5 Figure Generation (Questions 3 & 4)

Listing 5: Code for Figures 2 and 3.

```

1  # -- Figure: Q3 CI width comparison -----
2  r, sigma, T, M = 0.05, 0.30, 1, 200_000
3  K_list = [35, 40, 45, 50, 55]
4  widths_a, widths_g = [], []
5  for K in K_list:
6      _, sa = asian_option_mc(50, K, r, sigma, T, method="arithmetic")
7      _, sg = asian_option_mc(50, K, r, sigma, T, method="geometric")
8      widths_a.append(2 * 1.96 * sa / np.sqrt(M))
9      widths_g.append(2 * 1.96 * sg / np.sqrt(M))
10
11 x, w = np.arange(5), 0.35
12 fig, ax = plt.subplots(figsize=(7, 4))
13 ax.bar(x - w/2, widths_a, width=w, label="Arithmetic", alpha=0.85)
14 ax.bar(x + w/2, widths_g, width=w, label="Geometric", alpha=0.85)
15 ax.set_xticks(x); ax.set_xticklabels(K_list)
16 ax.set_xlabel("Strike $K$"); ax.set_ylabel("CI width")
17 ax.legend(); ax.grid(True, axis="y", linestyle="--", alpha=0.5)
18 plt.tight_layout()
19 plt.savefig("fig_q3_ci_length.pdf", bbox_inches="tight")
20
21 # -- Figure: Q4 price comparison -----
22 pa_list, pg_list, pbs_list = [], [], []
23 for K in K_list:
24     pa, _ = asian_option_mc(50, K, r, sigma, T, method="arithmetic")
25     pg, _ = asian_option_mc(50, K, r, sigma, T, method="geometric")
26     pa_list.append(pa); pg_list.append(pg)
27     pbs_list.append(CallBS(50, K, r, sigma, T))
28
29 fig, ax = plt.subplots(figsize=(7, 4))
30 ax.plot(K_list, pbs_list, "o-", label="European $C_{BS}$", linewidth=2)
31 ax.plot(K_list, pa_list, "s--", label="Asian (arithmetic)", linewidth=2)
32 ax.plot(K_list, pg_list, "^:", label="Asian (geometric)", linewidth=2)
33 ax.set_xlabel("Strike $K$"); ax.set_ylabel("Option price")
34 ax.legend(); ax.grid(True, linestyle="--", alpha=0.5)
35 plt.tight_layout()
36 plt.savefig("fig_q4_comparison.pdf", bbox_inches="tight")

```